

Lab02_Mazo

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Questão 1

Resposta:

Estatísticas suficientes:

n = quantidade de voos válidos; $n_{\text{atrasados}}$ = quantidade de voos com `ARRIVAL_DELAY > 10`

Percentual de atraso:

$\text{Percentual} = n_{\text{atrasados}} / n$

Questão 2

```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

`filter`, `lag`

Os seguintes objetos são mascarados por 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```

getStats <- function(input, pos) {

  input |>
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) |>
    filter(
      !is.na(ARRIVAL_DELAY),
      !is.na(DAY),
      !is.na(MONTH),
      !is.na(AIRLINE)
    ) |>
    group_by(DAY, MONTH, AIRLINE) |>
    summarise(
      n = n(),
      n_atrasados = sum(ARRIVAL_DELAY > 10),
      .groups = "drop"
    )
}

```

Questão 3

```

banco <- read_csv_chunked(
  "flights.csv.zip",
  callback = DataFrameCallback$new(getStats),
  chunk_size = 100000,
  col_types = cols_only(
    AIRLINE = col_character(),
    DAY = col_integer(),
    MONTH = col_integer(),
    ARRIVAL_DELAY = col_double()
  )
)

```

Questão 4:

```

computeStats <- function(stats_parciais) {

  stats_parciais |>
    group_by(DAY, MONTH, AIRLINE) |>

```

```

    summarise(
      n = sum(n),
      n_atrasados = sum(n_atrasados),
      .groups = "drop"
    ) |>
    mutate(
      Data = as.Date(
        paste(2015, MONTH, DAY, sep = "-")
      ),
      Perc = n_atrasados / n,
      Cia = AIRLINE
    ) |>
    select(Cia, Data, Perc)
  }

stats <- computeStats(banco)

```

Questão 5:

```
library(devtools)
```

Carregando pacotes exigidos: usethis

```
library(ggcal)
library(ggplot2)
```

```

pal <- scale_fill_gradient(
  low = "#4575b4",
  high = "#d73027"
)

baseCalendario <- function(stats, cia) {

  dados_cia <- stats |>
    filter(Cia == cia)

  ggcal(dados_cia$Data, dados_cia$Perc)
}

```

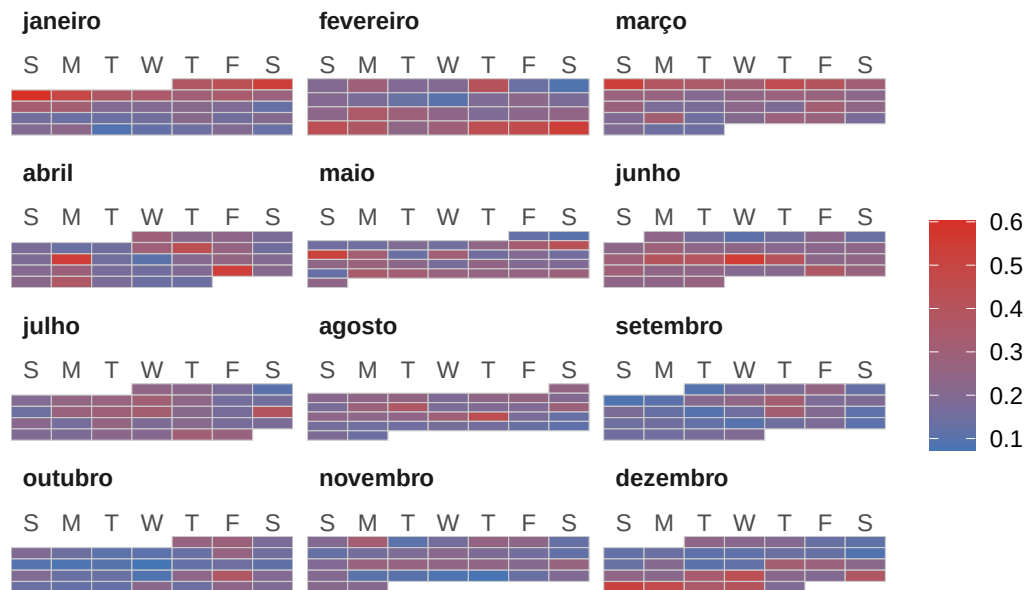
```

cAA <- baseCalendario(stats, "AA")
cDL <- baseCalendario(stats, "DL")
cUA <- baseCalendario(stats, "UA")
cUS <- baseCalendario(stats, "US")

cAA + pal + ggtitle("Atrasos - American Airlines (AA)")

```

Atrasos – American Airlines (AA)

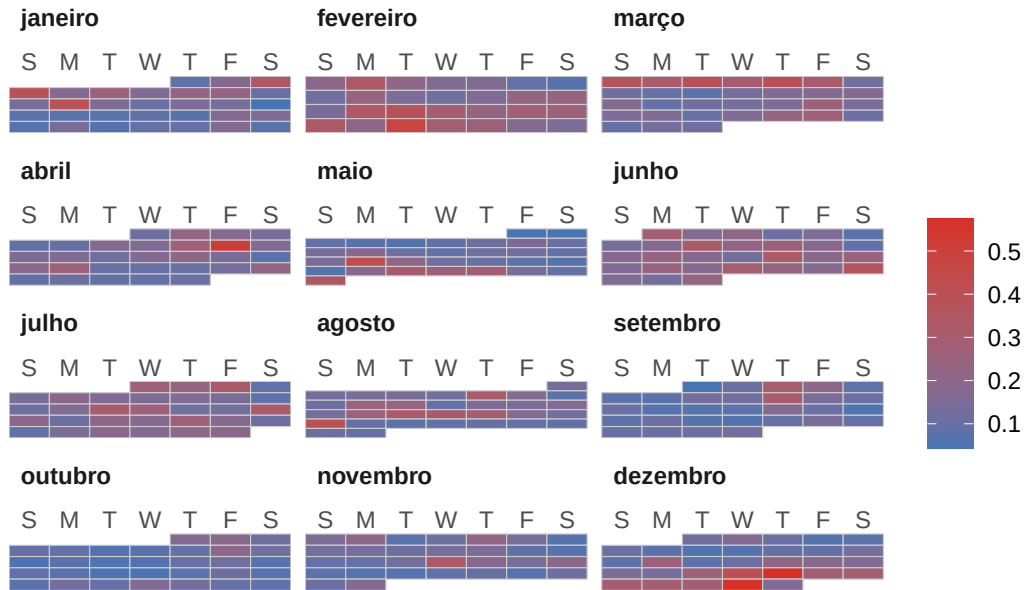


```

cDL + pal + ggtitle("Atrasos - Delta Air Lines (DL)")

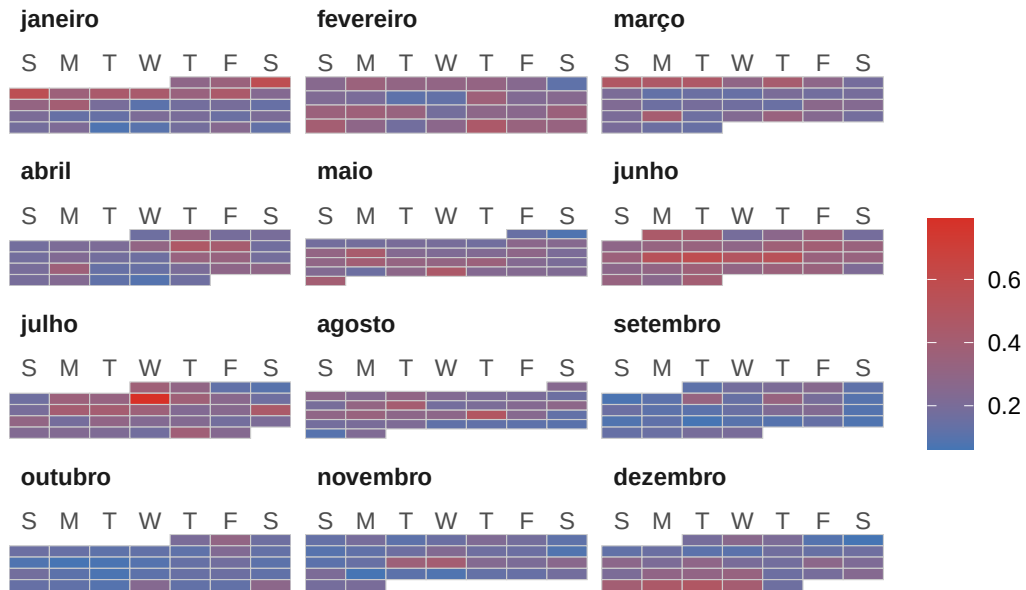
```

Atrasos – Delta Air Lines (DL)



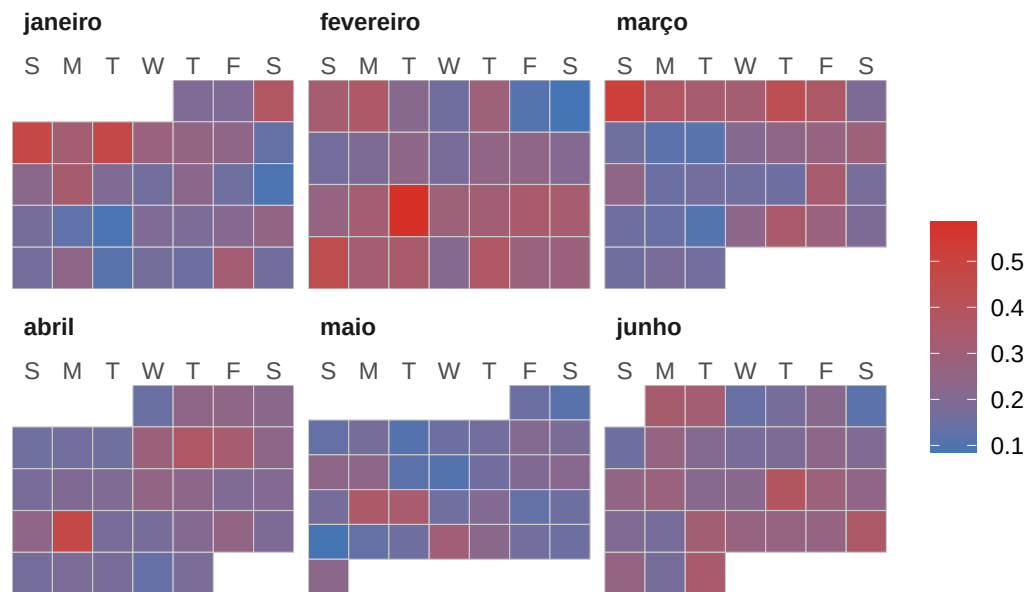
```
cUA + pal + ggtitle("Atrasos - United Airlines (UA)")
```

Atrasos – United Airlines (UA)



```
cUS + pal + ggtitle("Atrasos - US Airways (US)")
```

Atrasos - US Airways (US)



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-24 19:13:04 -03"
```